

# JONATHAN JONGBIN KIM

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## EDUCATION

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**Korea Advanced Institute of Science and Technology (KAIST)**

*B.S. in Electrical Engineering*

Daejeon, Republic of Korea

*Mar. 2025 – Present*

**GPA:** 3.65 / 4.3

**Relevant Coursework:** Introduction to Reinforcement Learning, Introduction to Generative Models, Machine Learning Basics and Practices, Probability and Introductory Random Processes, Linear Algebra

## RESEARCH INTERESTS

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Reinforcement Learning, World Models, Generative Models, Self-improving RL Agents

## SELECTED PROJECTS

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**3D Gaussian Splatting — Scene Reconstruction** [GitHub]

2026

*PyTorch renderer implementation and gsplat-based 3D reconstruction*

- Implemented a pure-PyTorch 3DGS renderer with 3D covariance, perspective projection, depth sorting, and differentiable alpha compositing.
- Built a COLMAP-to-gsplat training pipeline with SfM initialization, L1/SSIM losses, progressive spherical harmonics, and densification hooks.
- Added ablation controls for densification, parameter freezing, and scale/opacity regularization, with TensorBoard logging and PSNR/L1 validation.
- Trained on 135 registered views for 30,000 steps on a Tesla T4, achieving validation PSNR of 27.354 dB and L1 error of 0.03535 with 505,695 Gaussians.

**StarGAN v2 — Multi-Domain Image Translation** [GitHub]

2025

*Paper-based implementation and empirical evaluation*

- Implemented a PyTorch framework for  $256 \times 256$  multi-domain image translation on AFHQ, including a generator, mapping network, style encoder, and discriminator.
- Combined AdaIN-based style conditioning with adversarial, style reconstruction, diversity-sensitive, and cycle consistency losses in the training pipeline.
- Trained and evaluated the model on a MacBook Pro with an Apple M4 Pro and 24 GB RAM using the PyTorch MPS backend.
- Evaluated a 20,000-step checkpoint on 3,000 generated images, achieving an overall FID of 44.11 and LPIPS diversity of 0.4270; analyzed domain-specific generation quality and visual changes across training checkpoints.

## SELECTED PUBLICATIONS

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[1] **J. Kim**, et al., “Analyzing The Impact and Evaluation of Flexitime on Subway Congestion Using SARIMA Model”, *Proceedings of the Korea Computer Congress (KCC)*, 2024.

[2] **J. Kim**, et al., “Comparison of Deep Learning Architecture Performance and Suggestion of Baseline Model for Diagnosing Cassava Leaf Disease”, *Proceedings of the Korea Software Congress (KSC)*, 2024.

## HONORS & AWARDS

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**Excellence Award, Junior Division**

2024

Undergraduate/Junior Paper Competition, Korea Software Congress (KSC)

## SKILLS

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**Programming:** Python, C++, Bash

**Machine Learning:** PyTorch, NumPy

**Tools:** Git, Linux, CUDA, L<sup>A</sup>T<sub>E</sub>X